

Developing Intentional Relationships with Technologies: Exploring Players' Tension between Play and Non-play and Designing Built-in Interventions to Contribute to Digital Well-Being

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ABSTRACT

The dark side of technology, such as technology overuse, has received growing concern. Prior studies have developed a variety of stand-alone techniques to support technology non-use, but few have attempted to build interventions directly into technologies to regulate usage. My research seeks to address this issue in the context of games by exploring the relationship players intend to have with games, especially their desired balance between play and non-play, and designing built-in interventions in games to help support this relationship. I will conduct qualitative interviews to understand players' ideal relationships with games, and quantitative experiments to investigate the impact of built-in interventions on such relationships. Through these investigations, I attempt to develop a theoretical framework to understand users' intended relationship with technologies and provide guidance on intervention designs that contribute to digital well-being.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

intervention design, games, technology non-use, game experience, persuasive technology, built-in intervention

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1 CONTEXT AND MOTIVATION

As technology has rapidly developed, it has penetrated into people's lives. Despite the conveniences it brings, technologies also disrupt people's productivity [15], mental health [33], and social interaction [3, 23]. People are aware of these negative impacts, and

often attempt to restrict their technology usage [19, 28]. Among different types of usage, game playing is one of the most frequently mentioned activities that people seek to regulate [12, 20, 22].

Though gameplay brings a lot of enjoyment, many people struggle with excessive time spent on games. These people's social relationships, health, and performance at school or work are disrupted, and they experience negative feelings such as guilt [34]. This can also damage their relationship with games, as they feel burnout and get sick of the game after spending a significant amount of time in it [31]. Since games generally do not incorporate ways to help players become aware of the time spent playing, they have to rely on willpower or external strategies to counter excessive gameplay. In an attempt to cope, some people will even quit or delete the game [20, 31]. A large body of work, however, has demonstrated the difficulties associated with self-regulation [12, 15, 18], and the method of deleting the game may be neither satisfactory to the players nor to the game companies.

Prior studies have developed a range of intervention tools to support technology non-use, yet these interventions are generally stand-alone, separate from the targeted technology, and as a result have a number of drawbacks [7]. One major drawback is that a stand-alone intervention may be unable to support technology non-use in the context of real-time interaction, such as gameplay, that occurs *within* the activity that people want to regulate. To identify a more effective way to support technology non-use, I aim to understand the relationship people intend to have with technologies in the context of games, and investigate the design of built-in interventions in games to contribute to such a relationship.

2 RELATED WORK

Despite the long-standing focus on the use of technology, an increasing number of studies in the human-computer interaction (HCI) community has discussed the *non-use* of technology [2, 20, 26, 27]. This is largely due to the potential negative effect of technology. Previous research has highlighted that technology can harm people's work-life balance [32], mental health [33], social relationships [3], and the ability to think, create, and make judgment [5, 21].

Despite the dark side of technologies, people intend to have a healthy relationship with technologies, instead of being simply stopping their use of digital technologies. They are constantly negotiating engagement and disengagement with the technology rather than being stable users or non-users [26, 35]. For instance, while

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players may experience negative effects such as missing appointments because of losing track of time in games, they also find immersion in games helpful as it relieves boredom in certain circumstances [34]. Also, though people agree that the lock-out mechanism is the most effective way to support phone-related behavior change, they are reluctant to adopt this approach as they still value certain aspects of phone usage [30]. Such a nuanced balance between use and non-use suggests the need for work to understand the relationship people intend to have with technologies in greater depth.

Currently, a large body of work has investigated the tools to support technology non-use. These tools can be categorized into three groups: before, after, and during technology usage. Interventions that show up before technology usage prevent the user from accessing the technology according to certain user-defined rules [14, 17]. Interventions that appear after technology usage make use of self-monitor systems to enable users to reflect on their past usage and hopefully lead to behavior change [18, 24]. Interventions during technology usage are usually triggered while users are in the process of interacting with the technology, with some forms of “design friction” [8] or lock-out mechanisms [16].

Most of these tools, however, are built in stand-alone Apps or systems instead of being built directly into the targeted technology product. This type of stand-alone intervention fails to take the real-time human-computer interaction, such as what occurs in games, into consideration, possibly limiting its effectiveness. For instance, the impact of the stand-alone intervention has been shown to be less effective when applied to game playing [12]. This is probably because games are usually accompanied by goal-oriented tasks [9] and people are unwilling to stop gaming unless they reach these goals [11]. This indicates a need to take into consideration the ongoing player-game interaction when intervening in people’s game playing behaviors.

Likewise, Hiniker et al. [12] also suggested that researchers should consider building interventions directly into the technology that people are trying to regulate. Though technology companies, such as Google [10], and game companies, such as Nintendo [29], have taken the initiative to incorporate interventions into their products [13, 25], little academic research builds interventions directly into the targeted technology usage. In the context of games, such built-in interventions might be of great importance. This is because players are likely to immerse themselves in the game and even enter a state of flow, also known as “being in the zone”, while gaming [6]. Taking into account the real-time player-game interaction, built-in interventions might avoid breaking players out of immersion while still trying to address technology overuse.

Incorporating the work on technology non-use and tools to support technology non-use, my research seeks to understand the nuanced relationship people intend to have with technologies, especially their desired balance between use and non-use, and provides guidance on built-in interventions to support such a relationship. Specifically, I attempt to explore what I term an “intentional relationship” between users and technologies in the context of games, and design built-in interventions in games to foster this relationship.

3 RESEARCH QUESTIONS

To understand people’s experience of the tension between play and non-play and the relationship they intend to have with games, and to tailor the design of built-in interventions to that relationship, the research questions are set as follows:

RQ1: How do people experience the tension between play and non-play?

RQ2: What kind of relationships people intend to have with games?

RQ3: How might we design built-in interventions in games to support such relationships?

4 METHODS AND RESEARCH DESIGN

4.1 Study 1: Players’ Experiences with Built-in Interventions (Completed)

As people are having troubles with technology overuse [15, 33] and existing studies that address this issue mainly work on stand-alone interventions [14, 18], study 1 developed 3 kinds of built-in interventions aimed at reminding people to take a break in a casual mobile game. It sought to understand players’ experience, both positively and negatively, with these interventions. After the 4-day deployment study and the follow-up semi-structured interviews (N=16), I employed the thematic analysis to analyze the data following the steps suggested by Braun and Clarke [4].

Study 1 revealed unintended outcomes of built-in interventions and suggested players’ needs for interventions are context-dependent. This raised new questions such as what kind of relationship people intend to have with games and what are the contexts in which people intend to play games? These emergent questions prompted me to consider the concept of intentional relationship, which refers to the relationship people intend to have with technologies. As this concept needs further theoretical and operational definition, I plan to first develop an in-depth understanding of the relationships people intend to have with technologies before going into specific designs. Study 1 serves as a starting point for the following investigations into the concept of intentional relationships and design approaches that help people develop such relationships.

4.2 Study 2: Intentional Relationships with Technologies (In-Progress)

Though technology overuse can bring negative effects to people, people tend to strike a nuanced balance between use and non-use rather than being complete non-users. This suggests the need for a deeper understanding of players’ desired relationships with games. In study 2, I attempt to understand how people experience the tension between play and non-play and the relationship people intend to have with games through semi-structured interviews. Chinese players will be recruited as participants, as China leads the global game market in the number of players and revenue, and players in China spend the most time playing games [1]. I will ask participants questions with regard to their experience of the tension between play and non-play, their ideal relationship with games, the effect of playing games on them (e.g., life, health, work, study), the factors contributing to these effects, and the strategies they take to cope with these issues.

In this study, I use the grounded theory approach, a method that allows for thick description and rigorous qualitative data collection and analysis. With this approach, the earlier interviews will be as open-ended and participant-centered as possible to derive categories, and the later interviews will shift into a mutual conversation to develop categories. These categories will underscore the different dimensions of intentional relationships and will be analyzed to develop theories for intentional relationships.

4.3 Study 3: Operational Definition of Intentional Relationships (Planned)

In this study, I will attempt to provide a reliable and theoretically sound tool for researchers to measure players' intentional relationship with games. The theoretical definition of the intentional relationship will be based on the result of study 2. I will seek to operationally define intentional relationships through developing a scale, as the concept of intentional relationships might involve unobservable constructs and compose of different components.

The process of developing the scale includes item pools generation, questionnaire pilot test, exploratory factor analysis, and confirmatory factor analysis. The development of items will be based on (1) past research on technology non-use (2) qualitative analysis of interviews in study 2, (3) further focus groups to discuss the intentional relationship, if needed, and (4) quantitative analyses conducted in the current work. This scale will be administered to evaluate the intentional relationship between players and games.

4.4 Study 4: Built-in Interventions for Intentional Relationships (Planned)

After developing a deeper understanding of the relationships people intend to have with games, I plan to provide design guidance to support such relationships. As limited work explores built-in interventions, a potentially promising way to counter the dark side of technology, I will design built-in interventions in games and evaluate the effect of the interventions on the intentional relationship. The method of experiments will be used in this study, as it is beneficial in controlling confounding variables and drawing cause-and-effect conclusions. In the experiments, games with particular types of intervention designs will be created. The design of the built-in interventions will be based on the elements that emerged in study 1, such as the sense of accomplishment, negative affects, and embeddedness. Participants will be asked to play games incorporating these different intervention designs, and the impact of the designs will be evaluated through the scale developed in study 3.

5 RESEARCH PROGRESS

Study 1 has already been completed. The findings of study 1 showed that while some players had a positive experience with the game that had built-in interventions, others experienced unintended outcomes such as disrupted immersion, longer play sessions, and elevated mental stress. Players in this study also had contrasting opinions towards the embeddedness of interventions. Besides, the study suggested that interventions seemed to be more likely to succeed when players felt a sense of accomplishment or negative emotions, and that players' needs for built-in interventions were context-dependent.

Currently, I'm working on study 2. The interview guide of study 2 has already been developed. I'm in the process of recruiting participants and doing pilot tests of the guide. Study 2 is expected to be completed before January 2022. After that, I will proceed to carry out study 3 and study 4. By the time of the doctoral consortium, I am supposed to have completed study 2 and be working on study 3. I would love to share the results of study 1 and study 2 with mentors and my fellow participants during the event, and get feedback on the research design and data analysis techniques of the in-progress study 3 and the plan for study 4.

6 DISSERTATION STATUS AND LONG TERM GOALS

I am currently a third-year Ph.D. student at National University of Singapore. My advisors are Dr. Alex Mitchell and Dr. Renwen Zhang. I enrolled in my current Ph.D. programme in Aug 2019, and I passed my qualification exams in July 2021. I plan to carry out 4 studies for my dissertation. I have already completed study 1, and I'm currently working on study 2. Study 3 and 4 are expected to be completed. The projected completion date of my dissertation is Aug 2023. The maximum period for my Ph.D. candidature is 5 years.

Regarding my long-term goals in doing research, as my research interest lies in digital well-being and game studies, I intend to provide new insights into the relationship between human beings and technologies, especially between players and games, and provide guidance for technology design that promotes digital well-being. In terms of the long-term goal in my career, I hope to continue doing research in academia, especially in an institution that appreciates interdisciplinary research, which will allow me the opportunity to work with people from different areas and gain insights from different perspectives.

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